Operating Systems - Lecture 1
Instructor: Dr. Alan Turing

1. Introduction to Processes
A process is fundamentally an instance of a computer program being executed by one or many threads.
It contains the program code and its current activity. Depending on the operating system, a process may be mad

Key Components of a Process:
- Text Section: The compiled program code.
- Program Counter: Tracks the current address of the instruction being executed.
- Stack: Contains temporary data like function parameters, return addresses, and local variables.
- Heap: Memory dynamically allocated during process runtime.

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2. Virtual Memory & Paging
Virtual memory is a memory management technique that provides an idealized abstraction of the storage resour

How Paging Works:
The operating system divides memory into fixed-size blocks called pages (in virtual memory) and frames (in phys
The Page Table maps virtual page numbers (VPN) to physical frame numbers (PFN).
When a requested page is not in RAM, a 'Page Fault' occurs, prompting the OS to fetch the page from disk.